

Two formalizations of Scott’s approximation layer, compared

Ours — ScottDomains/, formalizing C. A. Gunter and D. S. Scott, *Semantic Domains* (HTCS Vol. B, 1990).

Theirs — scott1972/, Lars Warren Ericson, *A Lean 4 Formalization of Scott’s Continuous Lattices* (1972), arXiv **2606.30782**, formalizing D. S. Scott, *Continuous Lattices* (1972). Cloned at [../.. /scott1972](#); the paper is at [../papers/Ericson 2026 ... arXiv-2606.30782.pdf](#).

Both are sorry-free and both check on [propext, Classical.choice, Quot.sound]. Verified here rather than read off the abstracts: scripts/scott1972-verify.sh (936 jobs, exit 0, zero sorry in the sources) and scripts/scott1972-axioms.sh on Theorem 4.4’s two halves.

1. They are not the same subject

The two papers are eighteen years apart and sit on opposite sides of one distinction.

#		Scott 1972 (theirs)	Gunter & Scott 1990 (ours)
1	Carrier	complete lattice	cpo — directed sups, $\perp$, no top, no meets
2	Approximation	continuous : $y = \sqcup\{x \mid x \ll y\}$	algebraic : $y = \sqcup\{k \in K(D) \mid k \sqsubseteq y\}$, k compact
3	Entry point	topological — injective T_0 spaces, Scott-open sets	order-theoretic — directed sets and least upper bounds
4	$D \equiv [D \rightarrow D]$ via	the inverse limit D^∞	Theorem 21 + Lemma 23, no inverse limit at all
5	Countability	not required	required — Domain carries countable_compacts

Row 2 is a containment, not a disagreement: every algebraic lattice is a continuous lattice, and $x \ll x$ is exactly compactness. Row 4 is the sharpest divergence — PaperInventory.md records that §7 of the 1990 paper builds no inverse limit, and an early draft of that row listing D^∞ as an outstanding definition was struck for exactly that reason. **So this artifact formalizes the construction our paper deliberately declines to use.**

2. $\ll$ is defined differently in the two developments

Same symbol, same notation $\ll$, different definitions and different settings.

Theirs (Scott1972/ContinuousLattice/WayBelow.lean:72), topological, on a CompleteLattice:

```
def WayBelow (x y : D) : Prop :=
  ∃ U : Set D, ScottOpen U ∧ y ∈ U ∧ U ⊆ Set.Ici x
```

y lies in the interior of the principal up-set of x for the Scott topology, where ScottOpen U is IsUpperSet U plus inaccessibility by directed suprema.

Ours (ScottDomains/WayBelow.lean:51), order-theoretic, on a Preorder:

```
def WayBelow [Preorder α] (x y : α) : Prop :=
  ∀ (s : Set α) (u : α), s.Nonempty → DirectedOn (· ≤ ·) s → IsLUB s u → y ≤ u →
    ∃ z ∈ s, x ≤ z
```

Three consequences worth stating.

1. **Ours needs no suprema.** It is stated at [Preorder α] and quantifies over IsLUB rather than applying sSup, so it is available before any completeness is assumed. Theirs needs CompleteLattice for sSup and for the Scott topology to exist.
2. **Ours makes compactness definitional.** $x \ll x \leftrightarrow \text{IsCompactElement } x$ holds by Iff.rfl. Theirs has no compact elements at all — continuity, not algebraicity, is the 1972 notion.
3. **The two agree where both apply** (a complete lattice), by the standard equivalence between the topological and order-theoretic readings, but **neither development proves that** — and neither could import the other, see §5.

3. Size

#	Layer	Theirs	Ours
1	way-below / approximation	19 thms, 236 lines	7 thms, 117 lines
2	the approximation class	IsContinuousLattice, same file	13 thms, 227 lines — IsAlgebraic, Domain, BoundedComplete
3	function spaces	110 thms, 1633 lines	41 thms, 887 lines — ScottHom, StepFunction, FunctionSpaceDomain, CompactFunction, FunctionSpaceCountable
4	inverse limits / D^∞	47 + 33 thms, 1241 lines	none
5	whole development	273 theorems	1298 theorems

Row 3 is the striking one: their function-space layer is nearly twice ours in theorems and lines, for a smaller stated result. The reason is row 3 of §1 — working topologically means every continuity argument passes through Scott-open sets and the $\uparrow$ a basis, where ours works directly with directed sets and step functions.

Row 5 is not a quality comparison. Theirs formalizes 43 numbered results of one paper; ours formalizes 24 of 29 numbered results plus 91 prose claims of a different, longer paper, and carries powerdomains, bifinite domains and §7’s universal domains that Scott 1972 does not contain.

4. What each has that the other does not

Theirs, absent from ours:

- **Injective T_0 spaces** (Injective.lean) and the Sierpiński space as Prop — Scott’s topological characterization. We have no topology anywhere.
- **The Scott topology as a topology** — ScottOpen, closed under $\cap$ and $\bigcup_0$. Ours never constructs it; ExistingTheories.lean only #checks Mathlib’s Topology.IsScott without using it.
- **Inverse limits and D^∞** — InverseLimits.lean, FunctionSpaceTower.lean, and the capstone $D^\infty \equiv [D^\infty \rightarrow D^\infty]$.
- **The Milner correction** (MilnerCorrection.lean) — the March 1972 erratum without which Propositions 2.9, 2.10 and 3.3 are wrong as printed. A formalization catching a printed error is a pattern this project knows well: ours records the same for Lemma 9’s items 3 and 5, Theorem 16’s second conjunct, and three defects in §7.4’s printed pre-ordering.

Ours, absent from theirs:

- **Compact elements and algebraicity** — IsCompactElement, compacts, compactsBelow, and the countability condition that makes a Domain.
- **Powerdomains** — Hoare, Smyth and Plotkin, with Theorem 12’s initiality.
- **Bifinite domains**, the Plotkin order, and finitary projections.
- **§7’s universal domains** — Dyadic.U, Colimit.V, representability and p-representability.
- **Bounded completeness as a separate predicate**, which is what lets §6 drop it deliberately.

5. Why nothing can be shared mechanically

The artifact pins leanprover/lean4:v4.30.0; this project pins **v4.32.2**. They therefore carry two independent Mathlib checkouts — about 7 GB each, unable to share oleans — and **no import can cross between them**. Any transfer is a deliberate port of a definition or proof, read and rewritten, not an import.

Two further obstacles to reuse, both of which also apply to our own submission plans:

1. **Naming.** Both developments name after the source paper’s numbering — theirs Theorem212, ours thm11, lem23, prop15. Mathlib names describe the statement, so both would need the same rename before upstreaming.
2. **Carrier mismatch.** Their results are stated for CLat, a bundled complete lattice. Ours are stated for an unbundled [CompletePartialOrder α] with [IsAlgebraic α]/[Domain α] as separate classes. Porting a theorem means re-deriving it at the weaker carrier, not translating it.

6. What this is useful for

1. **A second opinion on $\ll$.** Two independent Lean definitions of the same relation, from two papers, is the strongest available check that ours says what it should. Proving them equivalent on a complete lattice would be a real result and is not in either development.

2. **The road not taken.** If the D_∞ construction is ever wanted here — it is not needed for Gunter & Scott, which is the point — this is a worked, kernel-checked route.
3. **A calibration for the audit.** r0038 measured our speculative-API rate at $\approx 3.5\%$. Running the same measurement over 273 theorems formalizing one paper would say whether that figure is a property of this project or of the genre.

7. Reproducing the measurements here

```
scripts/scott1972-verify.sh all      # cache fetch + lake build Scott1972
scripts/scott1972-axioms.sh          # #print axioms on Theorem 4.4's two halves
scripts/lean-decls.py --per-file <files>
```

All figures in this document come from those, run on 2026-08-08 against artifact HEAD and this project at commit a8474d3.